



The glial-enriched *Fabp7* gene regulates circadian electroshock seizure threshold and activity-dependent mRNA expression in mouse brain

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ABSTRACT

Background: Patients with epilepsy often experience increased frequency of seizures at night. Given the crucial role glial cells play in modulating neuronal excitability, we hypothesize that circadian changes in glia may affect changes in seizure threshold. Fatty acid binding protein 7 (*Fabp7*) is expressed in brain astrocytes and is involved in the transport of fatty acids, signal transduction, and gene transcription. Its mRNA expression levels rise and fall in a circadian rhythm and is necessary for normal sleep regulation. We examined if *Fabp7* influences electrically induced seizure threshold and differential gene expression in wild type (WT) vs. *Fabp7* knockout (KO) mice with and without seizure.

Methods: Male C57/BL6N (WT) and KO mice were maintained on a 12/12 hour light/dark cycle. Seizure thresholds were measured by administering a regularly increasing, once per day, electroshock stimulus. Seizure type, either general (GEST) or maximal (MEST) electroshock seizure threshold, was determined by the mouse's behavioral and motor response to the stimulus. Hippocampal/cortical tissue was taken at the time of MEST for mRNA-Seq / differential expression (DE) analysis.

Results: Results showed that KO mice required a higher current to elicit both GEST and MEST compared to WT mice. Both MEST and *Fabp7* KO led to remarkable changes in the transcriptome.

Conclusions: *Fabp7*, which is highly enriched in astrocytes, plays an integral role in modulating neuronal excitability. Mice lacking *Fabp7* have an increased seizure threshold to electrical stimuli, suggesting reduced excitability in brain regions where *Fabp7* is normally abundant. This is supported by our transcriptomic data as many activity induced genes have lower expression in *Fabp7* KO. Further work is needed to examine time-of-day effects and elucidate the pathway by which *Fabp7* alters seizure threshold and gene expression.

BACKGROUND

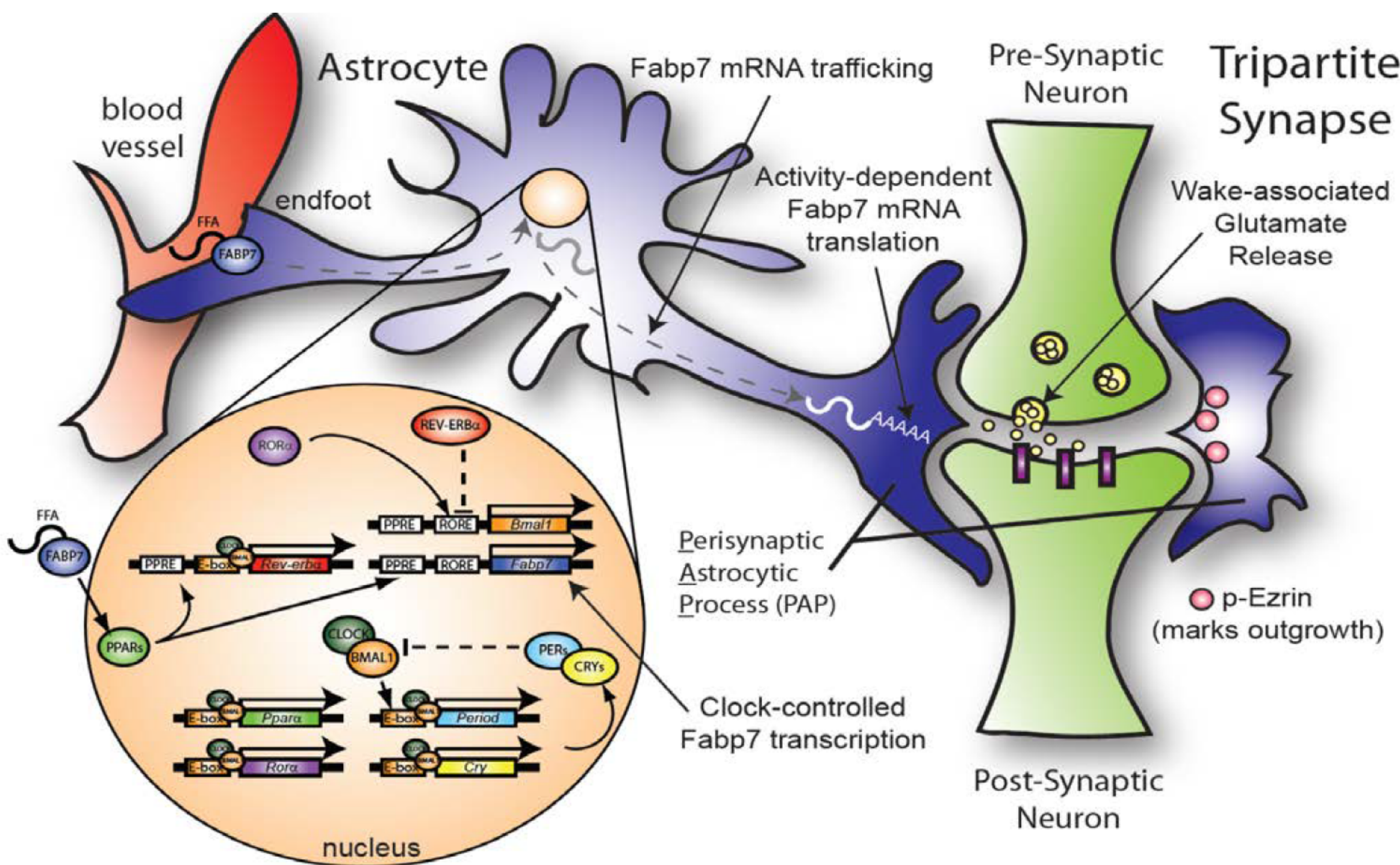


Figure 1. Glial *Fabp7* functional model. The tripartite synapse consists of a persynaptic astrocytic process (PAP) along with pre and post synaptic neuronal compartments. The expression of astrocytic *Fabp7* is regulated by the core circadian proteins *Bmal1* and *Rev-Erba*. Following transcription, *Fabp7* is trafficked to PAPs where it is translated upon behavioral state-dependent changes in neural activity. *Rev-Erba* is affected by electroconvulsive seizures and *Bmal1* has a role in the regulation of seizure susceptibility. It is therefore possible that expression of *Fabp7* contributes to epileptiform activity. We hypothesize that *Fabp7* functions to integrate neuronal excitability with circadian state.

Here we examined if electroconvulsive seizure threshold is altered in response to the presence or absence *Fabp7* and if there is a relationship to time-of-day. We also examined changes to the mRNA transcript profile caused by MEST and by *Fabp7* KO.

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RESULTS

General and Maximal Seizure Thresholds are higher in *Fabp7* KO mice during the dark phase

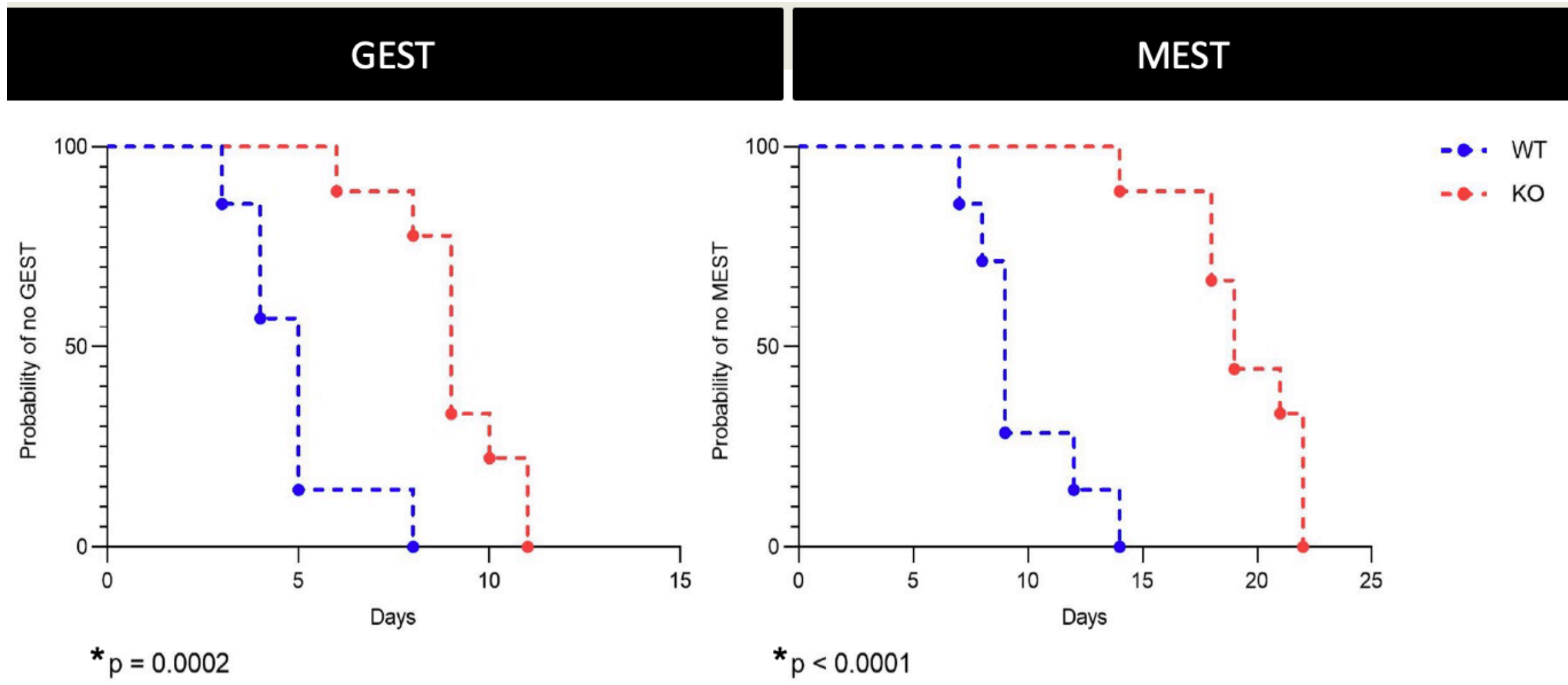


Figure 2. Seizure thresholds in WT and KO mice were measured across a 12:12 light:dark cycle, with no difference during the light phase (data not shown) and a difference during the dark phase (shown above). GEST was defined as the minimal current required to elicit a general tonic-clonic seizure with loss of upright posture and bilateral limb clonus. MEST was reached when the animal exhibited bilateral tonic hind limb extension. During the dark phase, *Fabp7* KO mice had higher GEST and MEST compared to WT ($p = 0.0025$ and $p < 0.0001$, post-hoc log-rank test).

Meta-Analysis reveals activity dependent genes are decreased by MEST in murine forebrain.

Sapkota et al., 2022			Hu et al., 2015		Current data	
Control vs Siezure			Control vs MEST		WT vs WT-M	
Log2FC	FDR		Log2FC	FDR	Log2FC	adj.P.Val
1.4747	9.16E-12	<i>Junb</i>	-0.7046	3.75E-11	-1.6593	1.29E-05
2.1836	1.99E-11	<i>Npas4</i>	-1.4793	8.32E-11	-2.9773	3.05E-08
1.0070	1.87E-08	<i>Btg2</i>	-0.8597	3.38E-05	-1.7356	3.54E-05
1.0846	2.36E-07	<i>Dusp1</i>	-0.9254	9.41E-24	-1.2023	1.72E-04
1.0380	8.46E-07	<i>Egr4</i>	0.7370	4.39E-04	-0.9139	1.85E-02
1.0269	1.62E-06	<i>Arc</i>	-0.6818	5.73E-03	-0.6159	1.38E-03
1.5597	1.12E-02	<i>Fosb</i>	-1.4793	4.14E-16	-1.0868	3.62E-03

Figure 3. RNA-Seq was performed on 5 samples of hippocampal/cortical tissue from each group: WT, WT-MEST, KO and KO_MEST at ZT20. Many Immediate Early-like transcripts were reduced in WT-MEST compared to WT-MEST as in Hu et al., 2015.

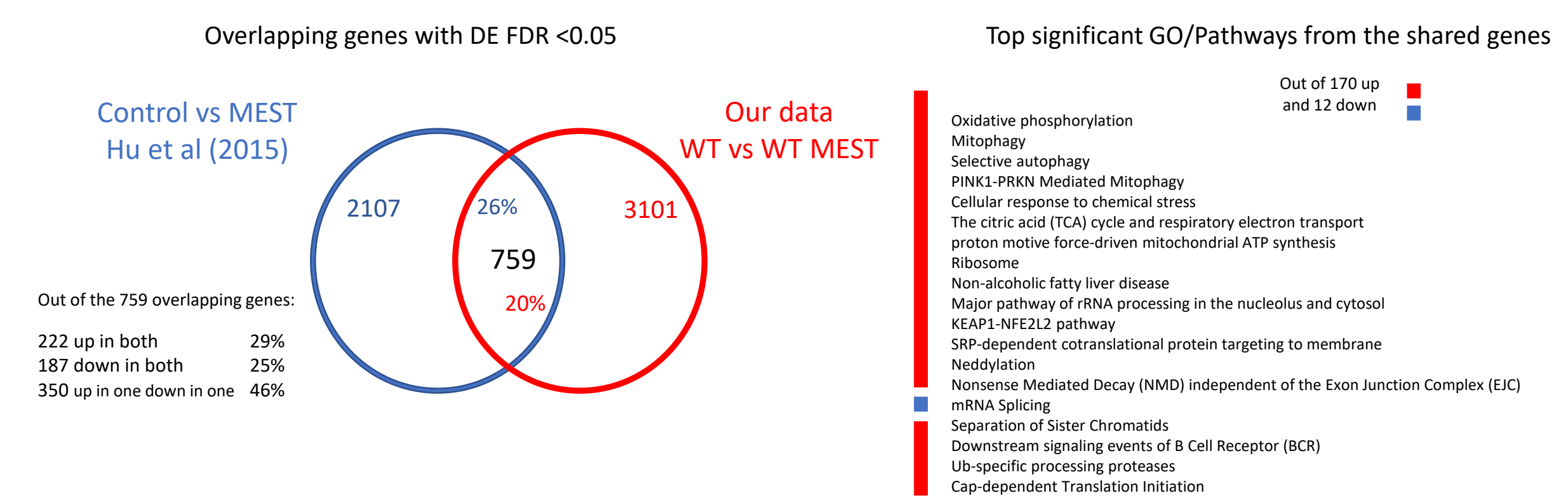


Figure 4. Other overlapping DE transcripts were also identified between studies. Approximately 10% of DE transcripts with FDR<0.05 are both shared and change in the same direction as the Hu et al. (2015) dataset.

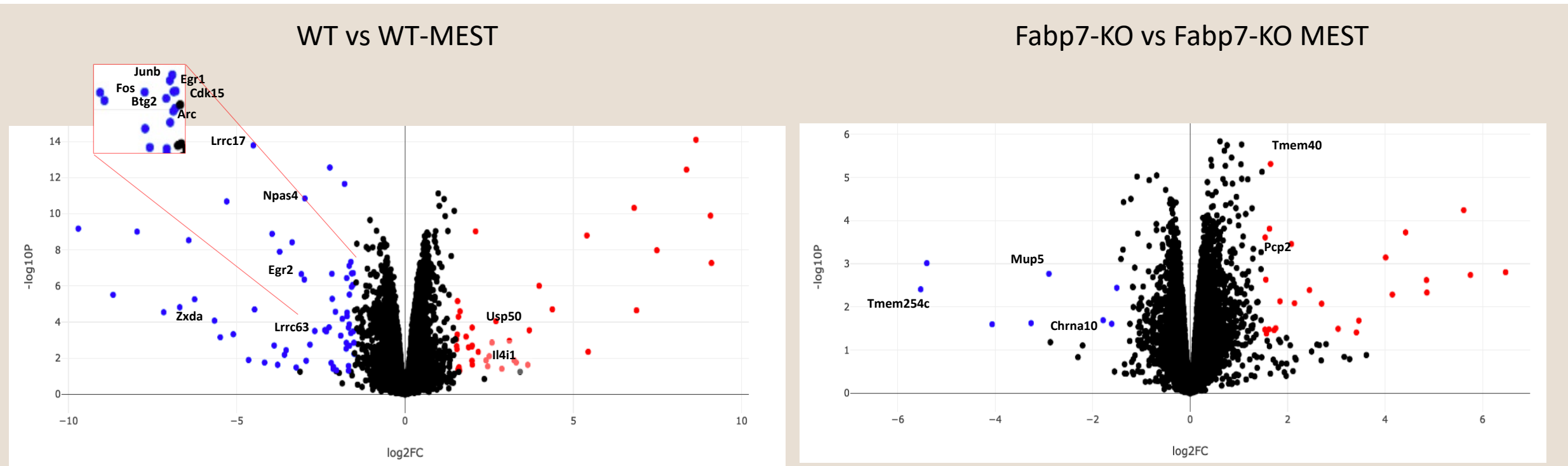
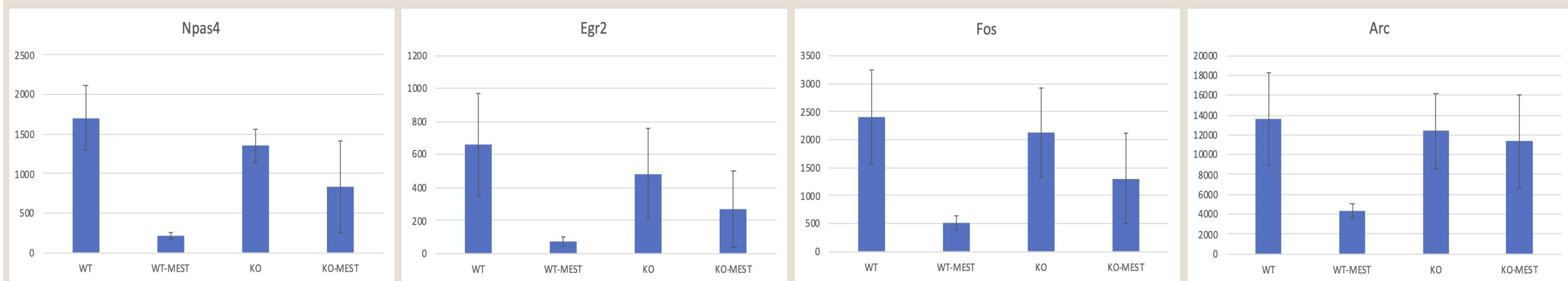


Figure 5. Volcano plots of DE genes from WT vs WT-MEST and KO vs KO-MEST demonstrate that *Fabp7* KO brains respond differently to seizure stimuli.

RESULTS

IEG expression has a different MEST response in *Fabp7* KO mice compared to WT mice



Normalized reads per million
Figure 6. IEG transcripts greatly decreased by MEST in WT, but not in *Fabp7* KO brain tissue. – stats...

Differential gene expression caused by MEST in WT is comparable to changes caused by *Fabp7* KO

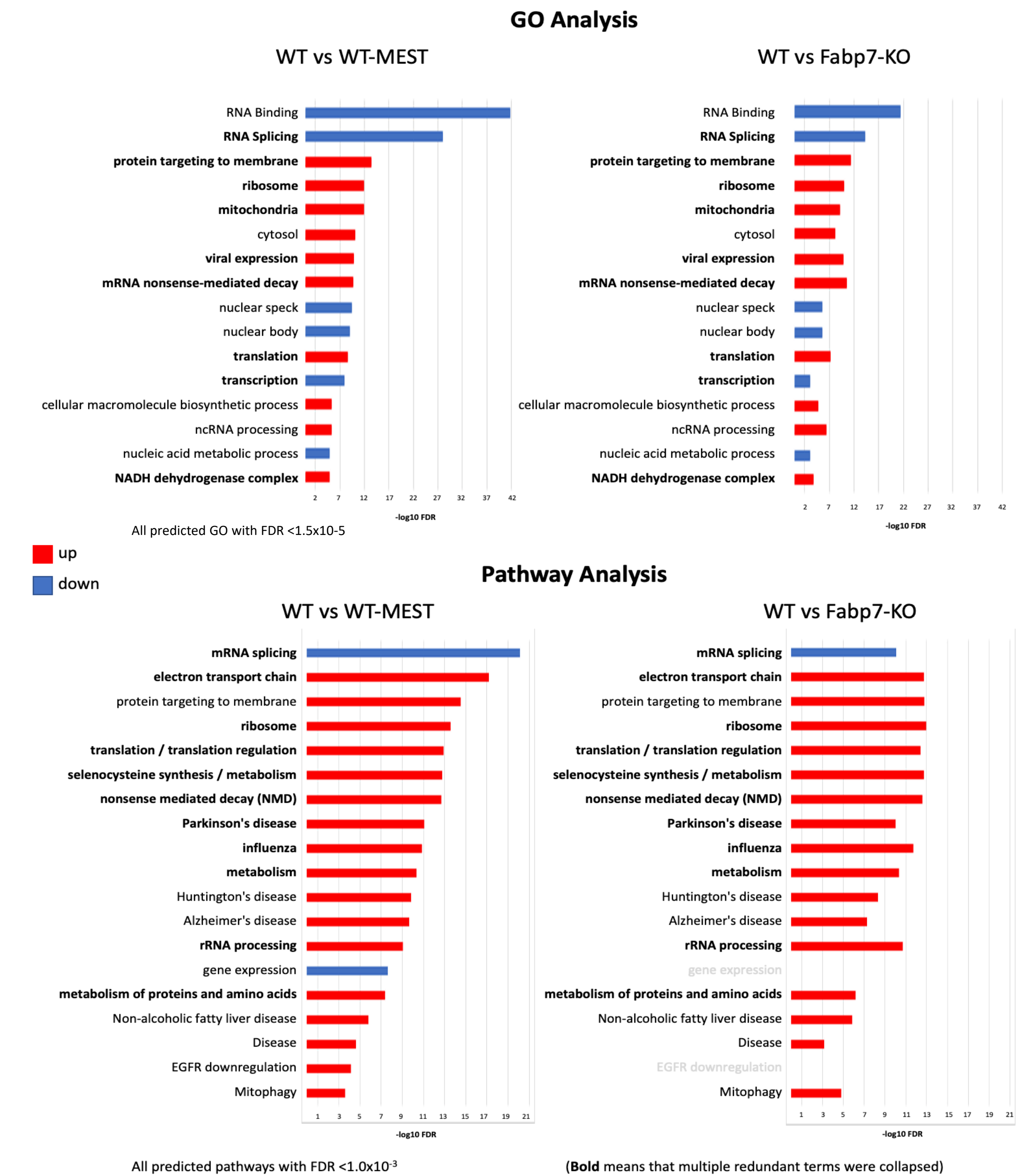


Figure 7. Comparison of WT to WT-MEST mRNA-Seq reads generated 3860 DE genes with an FDR<0.05, while the same comparison of KO vs KO-MEST returned only 201 genes (not shown). By contrast, in the absence of MEST, WT compared to *Fabp7* KO control SHAM have 2577 DE genes.

CONCLUSIONS

Fabp7 KO mice show time-of-day dependent increases in MEST

Wild-type MEST mice have a gene expression profile that is mimicked in the non-MEST *Fabp7* KO brain

Acknowledgments: NIH R35GM133440 "Characterizing evolutionarily conserved mechanisms underlying sleep, clocks, and memory"